

Insider Control, Group Affiliation and Earnings Management in Emerging Economies: Evidence from India

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Abstract

Using a sample of group affiliated and standalone firms for the years 2001-06 from India, a large emerging economy dominated by family business groups and firms with concentrated ownership, we examine the relationship between insider control and opportunistic earnings management with specific focus on the effect of business group affiliation on this relationship. We test for opportunistic earnings management in terms of the alignment and the entrenchment hypotheses and examine if such behavior is influenced by the complexity of ownership structures that are manifested in incomplete information on and fragmentation of declared ownership stakes. Our results provide evidence of a non-linear U-shaped relationship between insider control and opportunistic earnings management. This relationship holds true for both group affiliated and standalone firms though the marginal effects are stronger for the former. Finally, non-disclosure of complete ownership information and ownership fragmentation are found to be strongly related to opportunistic earnings management in group affiliated firms. Our results highlight that both insider control and group affiliation may independently influence agency costs in emerging economies. This in turn calls for policy actions that focus not only on individual firms but on business groups as consolidated identities.

JEL Classification: *G32, G34, G38, M41*

Keywords: *Corporate governance, ownership structures, regulation, earnings management*

1. Introduction

On January 7, 2009, Ramalinga Raju, the Chairman and controlling owner of Satyam Computer Services Limited – a blue chip infotech company in India as well as in the US, resigned from his post after publicly declaring that Satyam’s accounts have been manipulated and inflated for several years with his explicit involvement and without the knowledge of the board of directors.¹ The accounting fraud came into light after Satyam had to abort acquiring majority stakes in two family owned firms belonging to the Satyam Business Group and owned by Raju and his family members, due to pressure from institutional investors. Once the admission of fraudulent accounting came to light, the share value of the company plunged by nearly 80 percent in a matter of days.

Dubbed as the “Enron” of India, the Satyam case highlights how inside owners of family owned business groups with concentrated ownership and control can successfully extract and conceal private benefits at the expense of outside minority shareholders through earnings manipulation.² It is of interest to note that earnings manipulation by Satyam’s owners happened during a period much of which coincided with a steady increase in Satyam’s market valuation up until the acquisition announcements. The accounting irregularities were also reportedly perpetrated during a period that witnessed comprehensive governance regulations aimed at revising accounting standards to improve the integrity of financial statements.

The objective of this study is to examine in the context of India, a large emerging economy dominated by family business groups and firms with concentrated ownership, the relationship between insider control and opportunistic earnings management, with specific focus on the effect of business group affiliation on this relationship. Recent research using both country-level and cross-country analysis has shown that ownership structures by influencing the incentives of insiders, can have significant impact on the quality of financial reporting (see for instance, e.g., Fan and Wong, 2001; Haw et al., 2004; Ball and Shivakumar, 2005; Burgstahler et al., 2006). Extant research comparing corporate accounting frauds in the US and Europe, have also shown that differences in ownership structures broadly defined in terms of two prototypes – the widely held firm and the one with concentrated ownership and control-- can account for differences in the nature of accounting irregularities, the identity of those who commit these irregularities and the frequency of such incidents over time (Coffee, 2005).

¹ Raju publicly confessed that Satyam’s balance sheet as of September 30, 2008 contained inflated figures for cash and bank balances, had an accrued interest that did not exist, an understated liability and an overstated debtor’s position.

² The price of Satyam shares quadrupled between 2003 and 2005, and doubled between 2005 and 2006. Moreover, in September 2008, less than four months before the scandal broke, Satyam received the Golden Peacock Global Award for Excellence in Corporate Governance, instituted by World Council for Corporate Governance.

Despite the fact that the onus of several high profile corporate scandals both in developed and developing countries³ has been placed on concentrated ownership and complex family control structures, there are relatively few large sample studies, with even fewer country level studies, that have focused on examining the relationship between concentrated ownership and control and opportunistic earnings management (Bhaumik and Gregoriou, 2010). The need for such study is imperative in view of the widespread prevalence of concentrated ownership structures around the world and the fact that accounting failures have economy wide implications in terms of destabilizing the capital markets and reducing the credibility of financial reporting, both of which affect the ability of firms to access long term external finance. From a policy perspective, designing optimal financial reporting regulations and accounting standards by not taking into account the influence of insiders on the quality of financial reporting may lead to “unexpected or ineffective” policy outcomes (Leuz and Wysocki, 2008).

In contributing to the evidence on the effect of insider control on opportunistic earnings management in firms with concentrated ownership structures, we exploit detailed shareholding data on Indian firms to focus on two key relatively unexplored issues. The first issue involves examining in the context of firms in an emerging economy including those affiliated with business groups, the effect of insider control on opportunistic earnings management in terms of two competing hypotheses, namely the alignment and entrenchment hypotheses, with the former predicting that concentrated ownership would reduce the incentives of controlling insiders for opportunistic earnings management, while the latter predicting that such ownership will create incentives to engage in earnings manipulation to enable insiders to extract private benefits of control from minority shareholders. The second issue focuses on capturing how complexity of ownership structures arising from incomplete information and fragmentation of ownership stakes among inside owners that lead to ‘opaqueness’ of insider control, affect opportunistic earnings management.

With regard to the first issue, the few empirical studies that test the alignment and entrenchment hypotheses associated with concentrated ownership and opportunistic earnings management offer conflicting predictions. For example, Wang (2006) studies the role of the founding family in the US and finds evidence of a U-shaped relationship between family ownership and absolute discretionary accruals. The turning point occurs around 33 percent, suggesting that the negative entrenchment effect of family control outweighs the positive alignment effect fairly quickly as the extent of insider ownership increases.

³ While the Satyam accounting fraud can be cited as an example in the context of India, a developing country, the cases of Parmalat in Italy and Hollinger in Canada where controlling shareholders engaged in accounting irregularities and fraud can be cited as cases in the context of developed countries (Coffee, 2005).

In contrast, Ding et al. (2007) who study listed private sector firms in China find an inverted U shaped relationship with a turning point of 55 percent beyond which the alignment effect outweighs the entrenchment effect. The stark difference in results between these two studies suggests that the structure of concentrated ownership and control and the associated agency costs may matter in influencing the outcome. In particular, the US study focuses on a sample of large founding family firms with long term investors over multiple generations, whereas the sample for the Chinese study focuses on newly privatized firms which are not specifically family owned businesses. The institutional background of the countries may also matter as borne out by cross-country studies that suggest earnings management by ultimate owners may be significantly limited in countries with better statutory protection of minority shareholder rights (Luez et al., 2003; Haw et al., 2006). Our analysis of Indian firms includes two distinct types of firms namely those that are affiliated to family business groups and others that are essentially free standing entities. While the first category of firms are distinct from other ownership entities in terms of their ownership and control structures and the associated agency costs (Morck and Yeung, 2004), the second category are similar to firms in the US i.e. there is no cross-holding with other domestically listed firms in terms of ownership or control. We seek to contribute to the sparse and conflicting evidence by examining the competing hypotheses using this unique sample that captures the distinct attributes.

With respect to the second issue, we analyze the extent to which complexity in insider control structures facilitate expropriation of minority shareholders via earnings management. The complexity in concentrated ownership structures is often measured in terms of the divergence between control and cash flow rights of controlling shareholders. Referred to as the wedge, it is considered to be a major source of agency problem in firms in emerging markets. One common mechanism through which such divergence is created is through pyramiding of affiliated firms in business groups. The existing evidence suggests that higher wedge leads to greater opportunistic earnings management in emerging economies. Our analysis contributes to this literature by using a unique measure to proxy the pyramiding effect. In particular, we develop and examine the concept of ‘ownership opacity’ which arises largely due to the inability of outsiders to decipher the complete chain of insider ownership and control in listed firms. Using existing regulations on ownership disclosure, we identify two types of ownership opacity, one that arises due to incomplete disclosure of the identity of all insiders and their holdings, and the second arising due to fragmentation of insider ownership among multiple insiders.

The first type of opacity can arise if disclosure rules require the reporting of only those stock holdings that cross a particular threshold such as the 5 percent in the case of US companies, 25 percent in the case of

German listed companies, and one percent in the case of Indian listed companies. Such regulations can be strategically exploited by controlling shareholders, especially of family owned companies, to conceal the identity and extent of shareholding of insiders by deliberately keeping their ownership holdings at less than the regulatory threshold to avoid mandatory disclosures. In the case of India for example, the larger is the percentage of shareholding under the one percent cut-off, the less is the information available in the public domain, and hence more opaque is the ownership structure from the point of view of an outsider. We classify this as Type I opacity.

The second type of ownership opacity stems from the extent to which insider shareholding that is disclosed in the public domain is ‘fragmented’ among its declared constituents. This could be accomplished by distributing the disclosed ownership among a large number of inside owners with small ownership levels (just above one percent) rather than concentrating the ownership among a small number of large shareholdings. Many of these holders are generally unlisted companies and trusts on which further information, especially with their ownership structure, is not easily available in the public domain. Thus, by fragmenting, insiders can potentially create obstacles to active monitoring by outsiders by raising transaction costs. We term opacity arising from fragmentation as Type II opacity. We hypothesize that both types of opacity are engineered by the desire of camouflaging intra-group transactions and to facilitate earnings management for extracting private benefits.

Our focus on India is guided by three important considerations. First, poor quality financial reporting with high incidence of earnings management (Sarkar et al., 2008), the prevalence of concentrated ownership with the dominance of business groups characterized by pyramidal structures, and high incidence of related party transactions (Table 1) make India an ideal setting for exploring the relationship between insider control and earnings management. Expropriation of minority shareholders by controlling owners (referred to as promoters) is well documented in India both in official reports as well as in the business press⁴. At the same time, unlike other emerging economies, India ranks at par with the US and UK with respect to minority shareholder rights. However, similar to the other emerging economies, India lags behind the US and UK in terms of accounting standards and the rule of law.

Second, the richness of ownership data that has to be mandatorily disclosed under Clause 35 of the Listing Agreement of the country’s stock market regulator, Securities Exchange Board of India (SEBI),

⁴ For instance, apart from the Satyam, about 300 odd cases have been referred to the Serious Fraud Investigation Office in India since 2002. A large majority of these pertain to group companies and involve misreporting of company performance as well as misuse and diversion of investor funds by controlling promoters (Sarkar, 2010).

allows us to come up with a more accurate measure of insider control and compute measures of ownership opacity that has not been possible earlier.⁵ Specifically, a unique feature of Indian ownership data is that the disclosure of data on insider promoter ownership is based on the concept of control rather than on cash flows. Insiders have to disclose their total holdings in a company in terms of their direct cash flow rights in the company as well as their indirect cash flow rights in the company. The latter can be through companies they control and through individuals who act in concert with the insider. In contrast, in many of the existing studies on concentrated ownership, controlling insiders are identified by tracking down both their direct and indirect equity stakes through ‘equity chains’ using information available in the public domain and then defining different thresholds to define control (see for example, Claessens et al., 2000 and Lins (2003). Such an exercise may not be exhaustive due to missing data on all owners as is recognized in Claessens et al. (2000).⁶ In contrast, the mandatory disclosure of both direct and indirect ownership of *all* insiders in control in India helps to largely eliminate the “omission bias” that is in-built in many studies.

Insert Table 1 about here

Finally, the value of a study using India lies in the fact that comprehensive governance reforms to improve shareholder protection in line with international best practices have been introduced relatively early on. This allows us to examine the relationship between insider control and opportunistic earnings management in the backdrop of emerging governance regulations and the strengthening of the institutional framework over time.⁷ Corporate governance regulations for listed companies were implemented by the Securities Exchange Board of India (SEBI), the country’s capital market regulator, from January 2001 with the objective of instituting the highest standards of corporate governance in the country. The Clause specified mandatory standards with respect to almost all aspects of governance, both in terms of attributes and effectiveness, as it relates to the board, audit committee, investors grievance

⁵ Clause 35 of the Listing Agreement of SEBI requires insiders (traditionally referred to as “promoters”) to disclose on their own both their direct and indirect shareholding in the company through other persons/entities related to the insiders that together enables insiders to control the company.

⁶ For instance, Claessens et al. (2000), recognize the problem of omission bias in the context of computing ownership data and tracing the ultimate owner in East Asian corporations; data for 1164 corporations of the 5284 corporations considered, were missing or insufficient. Lins (2003), analyzing ownership structures in a cross-section of emerging economies had to eliminate China and Poland from the sample on account of not being able to identify 90 percent of the blockholdings in half the sample firms.

⁷ The Clause was made applicable in a phased manner and all listed companies with the paid up capital of INR 30 million and above or net worth of INR 250 million more at any time in the history of the company, were covered as of March 31, 2003.

committee, remuneration committee as well as disclosures of ownership patterns, related party transactions and CEO/CFO certification of compliance with all the provisions of this Clause. Companies were required to file a compliance report every quarter and include a “Corporate Governance Report” in their annual reports.

Using a sample of listed group affiliated and non-affiliated (stand alone) companies for the years 2001-06, and comprising of 5888 firm year observations, our results provide evidence of a non-linear relationship between insider control and opportunistic earnings management as measured by absolute discretionary accruals – insider control reduces income smoothing up to a threshold, beyond which the entrenchment effect associated with concentrated ownership outweighs the positive alignment effect and increases income smoothing. Put differently, insider control beyond a threshold level creates stronger incentives to manage earnings opportunistically to extract private benefits of control. This U-shaped relationship, between insider control and absolute discretionary accruals, also holds separately for group and non-group firms although the marginal effects are different. Additionally, consistent with existing studies, we find group affiliation to have an adverse effect on opportunistic earnings management. In particular, when insider control exceeds a threshold level of 26 percent, group affiliation has an adverse effect on earnings management. Finally, we find that ownership opacity exerts a significant effect on earnings management. Consistent with our expectations, we find that higher ownership opacity is associated with higher levels of absolute discretionary accruals only in the case of group affiliates. However, the effect of opacity is found to be sensitive to the type of opacity and the extent of insider control.

The paper is organized as follows. Section 2 discusses the background literature on insider control and earnings management. Section 3 discusses the sample of our study and defines the relevant variables. Section 4 discusses the empirical models estimated to examine the impact of insider control and ownership opacity on opportunistic earnings management. Sections 5 and 6 present the main empirical results of our analysis with respect to insider control and opacity, respectively. Section 7 concludes the paper.

2. Background Literature

The theoretical and empirical literature on concentrated ownership and opportunistic earnings management reveals conflicting predictions in terms of the alignment and entrenchment hypotheses. In Asian economies including India, concentrated ownership and control is the rule than the exception. When ownership and control are concentrated in the same hands, the nature of the agency problem in

firms changes vis-à-vis diffuse ownership structures, from shareholder-manager conflicts (Type I or vertical agency problems) to conflicts between two categories of principals – the controlling inside shareholders and minority outside shareholders who are typically dispersed (Type II or “horizontal agency problems”) (Roe, 2004). The basis of the alignment hypothesis is that the presence of concentrated insider ownership and control can help align the interests of inside management with that of diffused outside shareholders and thus mitigate Type I agency problems. This is achieved if the objectives of both insiders and outsiders are defined in terms of firm value maximization (Morck et al., 1988; Anderson and Reeb, 2003; Demsetz and Lehn, 1985). If earnings management is perceived to be detrimental to firm value, the alignment hypothesis implies that insiders with concentrated equity positions will have stronger incentives to control opportunistic earnings management by managers. If they are key members of management, it is expected that they would refrain from engaging in such opportunistic behavior if the alignment hypothesis is valid. The second premise of the alignment hypothesis particularly for family owned firms is that insider control, when synonymous with family control, can create incentives to report earnings in ‘good faith’ so as to prevent damage to the family reputation and protect family wealth in the long run over future generations (Wang, 2006). Reputational considerations are particularly considered strong in case of family firms that seek to tap external sources of finance at favorable contracting terms. This may be jeopardized by insiders trying to generate gains in the short term through opportunistically managing earnings.

The entrenchment hypothesis focuses on the conflict between the controlling insider shareholders and minority outside shareholder (Type II agency problems). It highlights how the entrenched controlling shareholders extract private benefits of control at the expense of minority shareholders (Shultz, 1988; Gilson and Gordon, 2003). Controlling shareholders can become entrenched due to several reasons. Their concentrated ownership positions can make them less susceptible to external disciplinary forces from the takeover market. Also when they have control rights in excess of their cash flow rights or when the equity structure is complex as is largely the case with business groups, they can become entrenched. Further, their presence in company boards, often in the capacity of Chairman and/or CEO (Ali et al., 2007) can also lead to entrenchment.

Under the entrenchment hypothesis, the motivation of controlling shareholders to engage in opportunistic earnings management stems from their need to camouflage the different ways in which they seek to acquire and retain private benefits of control. For instance, controlling shareholders can manipulate a firm’s earnings to extract private benefits of control through expropriation of cash, assets and equity (Bhaumik and Gregoriou, 2010). The ability of controlling owners to engage in earnings manipulation

and prevent subsequent detection stems from their absolute control of the firm including the generation and reporting of accounting information (Fan and Wong, 2002). With insiders typically present on company boards, the board of directors as a whole may be ineffective in monitoring and preventing insiders from engaging in such activities (Wang, 2006; Han An and Naughton, 2004).

While the benefits and costs associated with the alignment and entrenchment hypothesis as outlined above generally apply to firms with concentrated ownership and control such as family owned firms, additional costs and benefits come into play when such firms are associated with business groups which have been an “important and a persistent phenomenon” in many countries around the world (Masulis et al., 2009). Typical features of business groups are equity interlinks among group affiliates through pyramidal structure and cross-holdings which allow the ultimate owner of a group to control group firms without committing much equity. Additionally, group affiliates are connected through internal capital and labor markets, through related party transactions and through implicit relation based ties. On the benefits side, under the alignment hypothesis, the reputational effects associated with earnings management are likely to be stronger for business group firms interlinked through internal capital and labor markets, because potentially adverse effects of earnings management in one firm could have negative spillovers on other group firms (Gopalan et al., 2007). This would create even stronger incentives for the ultimate owner to ensure that group affiliates report earnings in “good faith.” Group reputation is particularly argued to be beneficial in emerging economies where minority investors are more vulnerable to expropriation due to weak regulatory and legal environment (Khanna and Yafeh, 2007).

On the costs side, internal markets of business groups, coupled with pyramidal structures and related party transactions, can strengthen the entrenchment effect associated with earnings management by enabling the manipulation, distortion and distribution of group profits and earnings estimates across affiliates. As Kim and Yi (2005) argue in the context of Korean business groups, business group affiliation provides insiders with more “opportunities and means” to use internal capital markets and related party transactions to engage in earnings management through asset acquisition and sales, equity investment, credit purchases and sales, payables and receivables, and in general through adjusting the volume or the price of intra-group trade. Empirical evidence exists for instance on how controlling shareholders in pyramidal group structures with related party transactions have engaged in earnings management to divert resources from over-performing group affiliates where controlling insiders have lower equity stake to ‘prop up’ underperforming group affiliates where insiders have higher stakes (Jian and Wong, 2004; Liu and Lu, 2004).

As pointed out in the introduction, studies focused on estimating the alignment and entrenchment effects of insider control on opportunistic earnings management are confined to the one with respect to US family firms and the other with respect to Chinese listed firms. While there are several studies examining the effect of inside ownership in firms with concentrated ownership, most of these are with respect to family firms in the US and are specifically confined to comparing whether family firms engage in higher earnings management compared to non-family firms. The general approach of these empirical studies has been to use a dummy variable proxying for family ownership and estimating its effect with respect to earnings management as measured by discretionary accruals. The implicit assumption underlying this specification is that the effect of family ownership may “go beyond” its ownership as measured by the extent of family shareholding. Using this approach, the dominant finding in this literature is that family ownership is associated with lower discretionary accruals as compared to non-family firms. The implicit conclusion drawn from such results is that the benefits of family ownership in mitigating Type I agency problems (alignment effect) outweighs the costs of family ownership in terms of extracting private benefits of control at the expense of minority shareholders (the entrenchment effect). However, on explicitly considering the extent of family ownership in terms of the magnitude of equity ownership by family members in US firms, Wang (2006) finds evidence of a U-shaped relationship between founding family ownership and discretionary accruals indicating that entrenchment effects do set in at higher levels of family ownership, beyond 33 per cent. Further, contrary to most findings, the dominance of family firms vis-à-vis non-family firms in Wang’s (2006) study are not unmitigated; the former report lower discretionary accruals only up to a particular threshold, of 67.4 per cent.

Studies on insider control and earnings management in emerging economies have focused on the effect of divergence of control rights and cash flow rights (wedge) on discretionary accruals. Country level analysis of wedge is restricted to an analysis of Korean listed firms (Kim and Yi, 2005; Han and Naughton, 2004) and cross-country analysis of firms with concentrated ownership and control in select East Asian and European countries (Haw et al., 2006). The Korean evidence is mixed: Kim and Yi (2005) find a higher wedge (implying higher incentives to extract private benefits of control) to be associated with higher opportunistic earnings management while Han and Naughton (2004) find no statistically significant effect. The cross-country studies do find evidence of an adverse effect of higher wedge on discretionary accruals.⁸ A few emerging economy studies have focused on the comparison of earnings management phenomenon in group affiliated vis-à-vis stand alone firms. Both Kim and Yi (2005), using

⁸ Fan and Wong (2002) too estimate in the context of select East Asian countries, the effect of wedge on earnings quality but focuses on earnings informativeness rather than earnings management.

Korean data, and Siregar and Utama (2008), using Indonesian data, find that group affiliated firms engage in higher earnings management relative to non-affiliated firms. Finally, research examining the effect of large blockholding on earnings management in a sample of 142 privately listed Chinese companies, some of which are family owned (Ding et al., 2007),⁹ shows that the relationship between the holdings of the largest shareholding block and discretionary accruals is an inverted U with higher concentration of shareholdings of the largest shareholders beyond a threshold of 55 percent leads to lower earnings management. This result, as mentioned in the introduction, is contrary to the U-shaped relationship associated with concentrated ownership and control in family firms in the US.

The difference in the results between Wang's (2006) analysis of US founding family firms and Ding et al.'s (2007) study of private listed Chinese firms could very well be due to sample characteristics, especially with respect to the type of concentrated ownership. It could also result from differences in the institutional set up in which the firms operate. The importance of the latter is often highlighted in studies of ownership structures and agency problems in emerging economies. The main argument is that while concentrated ownership structures have come about in emerging economies as an optimal response to solving Type I agency given the weak legal and regulatory institutions, weak minority investor rights, and weak enforcement mechanisms, weakness of these institutions have at the same time aggravated Type II agency problems (see, Berglof and Claessens, 2006; Peng et al., 2008). Hence, it is argued that relationships that hold, say with respect to ownership and firm value in developed countries such as the US with "well regulated and transparent markets" may not necessarily hold for emerging economies such as East Asian countries (Anderson and Reeb, 2003; Fan and Wong, 2002). Evidence of this in the specific context of ownership structure and earnings management is provided by Haw et al. (2004) who find that the extent to which the wedge between control and cash flow rights adversely affects discretionary accruals is significantly limited in countries with stronger protection of minority investor rights.

There are some similarities and differences between our study and those using Chinese data (e.g. Ding et al, 2007). Similar to the study on China, our analysis focuses on listed firms in an emerging economy. However, unlike the China based study, our sample is much more representative of firms in other emerging economies. Our sample helps understand the behavior of family owned and controlled firms,¹⁰

⁹ A listed company in China is defined as in which the largest shareholder is a private company or an individual.

¹⁰ Based on a sample of 307 large listed firms among the top 500 firms ranked in terms of market capitalization, it is found that as of March 2006, the proportion of family insiders as directors on company boards in group affiliates and standalones are 77 percent and 75 percent, respectively.

as well as the effect of business group affiliation on pyramidal structures and cross-holdings.¹¹ Corporate governance institutions in India are well defined and at par with those in the US and UK (Sarkar and Sarkar, 2008), have evolved over a sufficiently long time¹², are well-defined, stable and are relatively advanced with respect to the range and depth of existing statutes and legal framework regulating corporate activities (Sarkar and Sarkar, 2000). This is in stark contrast to other emerging economies, where corporate governance institutions are relatively new. India, however, like most emerging economies, have an enforcement regime that is weak relative to the “laws in the books.” Several recent research have argued that good quality enforcement is as important as good quality laws in order to reduce agency costs.

3. Sample and Variables

3.1 Sample

The sample for our analysis is drawn from the Prowess database created by the Center for Monitoring Indian Economy (CMIE). The Prowess database has formed the basis of several empirical studies on the Indian corporate sector (see for example, Khanna and Palepu, 2000; Sarkar and Sarkar, 2000; Bertrand et al., 2002). Our sample consists of all domestic private sector companies listed on the Bombay Stock Exchange for which complete accounting and stock market data are available for the financial years 2002 to 2006.¹³ We focus on listed companies as most of the recent capital market regulations including Clause 49, are applicable to only listed companies. From this set we focus on domestic private sector companies as the issues relating to promoter ownership and pyramidal structure are mostly applicable to domestic companies.

--- Insert Table 2 here ---

¹¹Compared to Indian private sector listed firms, Chinese listed firms are relatively young with the first privately owned listed company appearing in 1992, and accounting for only 16 percent of the total listed firm in the Chinese stock market as of 2002 compared to more than 80 percent of the total listed companies in the Indian stock market. Further, in considering the ownership of the top shareholders, the analysis of Chinese listed firms considers only direct ownership and indirect ownership through pyramids is outside the scope of the study.

¹² For instance, the history of capital market in India dates back to more than hundred years with the establishment of the Bombay Stock Exchange in 1875, and the origins of the existing Companies Act, 1956, which governs the activities of Indian companies has its roots in the Indian Companies Act of 1913.

¹³ The financial year in India is from April 1 to March 31 of the following year. Thus the financial year 2003 covers the period from April 1 2002 to March 31 2003.

To select our sample, in Step 1 we consider those listed companies in Bombay Stock Exchange which satisfy the following five criteria, namely, (i) they belong to non-agricultural industries, (ii) have accounting and market data available for the financial years 2002 to 2006, (iii) are not government or foreign owned, (iv) are not classified in the Z category by exchange¹⁴ (iv) and do not have outliers with regard to variables that capture firm characteristics (v). In Step 2 we delete those companies that have missing equity ownership information, and finally in Step 3 we delete those companies with missing information for computing discretionary accruals (which requires a minimum number of observations per industry), earnings response coefficients and control variables. Our final sample consists of an unbalanced panel of 1658 distinct companies with 5888 company-year observations. Of these observations, 2881 observations (48.93 percent) come from group affiliated companies and the remaining from unaffiliated or standalone companies; similarly, 4614 observations (78.36 percent) are following the adoption of Clause 49 and the remaining are from the pre-regulation period. Details of the sample selection procedure are presented in Table 2.

3.2 Dependent Variable: Opportunistic Earnings Management

Earnings management can be accomplished by the choice of accounting methods and by the assumptions and estimates used in computing accruals. It is easier to manage earnings using assumptions and estimates since the choice of accounting methods has to be disclosed under a country's accounting standards. Such earnings management is referred to as opportunistic earnings management in the literature. Healy and Wahlen (1999) define earnings management as being essentially opportunistic. Change in total accruals over time can proxy for the extent of earnings management if we assume that some portion of accruals is non-manipulated and constant over time. However, as suggested by Jones (1991), the non-manipulated accruals may not be constant over time and be driven by some accounting fundamentals such as revenues adjusted for receivables, property, plant and equipment. Opportunistic earnings management can be then be captured by the residuals from a regression of total accruals on accounting fundamentals. The residuals are referred to as discretionary accruals.

Further modifications to estimating discretionary accruals have been suggested in the literature since Jones (1991). Dechow et al. (1995) suggest that discretionary accruals be calculated after adjusting for change in receivables since it is often easier to exercise discretion over credit sales than cash sales.

¹⁴ The 'Z' group was introduced by BSE in July 1999 and includes companies which have failed to comply with its listing requirements and/or have failed to resolve investor complaints and/or have not made the required arrangements with both the depositories, viz., Central Depository Services (I) Ltd. (CDSL) and National Securities Depository Ltd. (NSDL) for dematerialization of their securities. See http://www.bseindia.com/about/list_comp.asp for details.

Kaszniak (1999) suggests controlling for return on assets in estimation of discretionary accruals whereas Kothari et al. (2005) suggest adjusting firm level discretionary accruals by that of a performance matched portfolio. Dechow and Dichev (2002) introduce cash flow measures to assess accruals quality. Finally, Francis et al. (2005) combine the model of Dechow et al. (2002) and Jones (1991) to construct an encompassing model of earnings quality. Since the Francis model is an encompassing model of discretionary accruals, we use the Francis et al. (2005) model to proxy earnings management in our analysis. The Francis model for estimating discretionary accruals is given by:

$$\begin{aligned} TCA_{it} / A_{i,t-1} = & \alpha_1(1/A_{i,t-1}) + \alpha_2(\Delta REV_{it})/A_{i,t-1} + \alpha_3(PPE_{it})/A_{i,t-1} + \alpha_4(CFO_{it-1})/A_{i,t-1} \\ & + \alpha_5(CFO_{it})/A_{i,t-1} + \alpha_6(CFO_{it+1})/A_{i,t-1} + \varepsilon_{it} \end{aligned} \quad (1)$$

where, CFO_{it-1} , CFO_{it} and CFO_{it+1} are the cash flows from operations in the years (t-1), t and (t+1), respectively; ΔREV_{it} is the change in revenues, PPE_{it} is the change in property, plant and equipment, and $A_{i,t-1}$ is total assets and ε_{it} is the error term. The dependent variable, TCA_{it} , is the total current accruals in year t calculated as $TCA_{it} = (\Delta CA_{it} - \Delta CL_{it} - \Delta Cash_{it} + \Delta STD_{it})$. Here ΔCA_{it} is the change in current assets, ΔCL_{it} is the change in current liabilities, $\Delta Cash_{it}$ is the change in cash and cash equivalents, ΔSTD_{it} is the change in short term debt in current liabilities.

The parameters of equation (1) are estimated for each year and each industry in our sample. Using the National Industrial Classification (NIC)¹⁵ code at the two-digit level, the sample companies can be classified into 25 different industries. We require that there are at least 10 observations in each industry in each year to ensure we have representative parameter estimates. Discretionary accruals (DA) are then calculated as:

$$\begin{aligned} DA = & TCA_{it} / A_{i,t-1} - [\hat{\alpha}_1(1/A_{i,t-1}) + \hat{\alpha}_2(\Delta REV_{it})/A_{i,t-1} + \hat{\alpha}_3(PPE_{it})/A_{i,t-1} + \hat{\alpha}_4(CFO_{it-1})/A_{i,t-1} \\ & + \hat{\alpha}_5(CFO_{it})/A_{i,t-1} + \hat{\alpha}_6(CFO_{it+1})/A_{i,t-1}] \end{aligned} \quad (2)$$

where, $\hat{\alpha}_1 - \hat{\alpha}_6$ are the parameter estimates from equation (1).

¹⁵ The National Industrial Classification (NIC) code is published by the Central Statistical Organization, New Delhi, India, and has undergone several revisions since its inception in 1970. We use the NIC-1998 classification which is identical to the International Standard Industrial Classification (ISIC) – Rev. 3 in structure up to the 4 digit level.

This measure identifies discretionary accruals as those that have a low correlation with realized cash flows. Francis et al. (2005) also measure accruals quality using standard deviation of the residuals from equation (1) over the previous five years and further separate it into innate and discretionary components. Given the long historical data requirements for this approach, we focus on examining the level of discretionary accruals using their model.

3.3 Variables of Interest

Our main variables of interest are equity ownership of controlling owners (henceforth, promoters), group affiliation, and ownership opacity.

3.3.1 Insider Control and Group Affiliation

Clause 35 of the Listing Agreement introduced by SEBI in 2002 provides the format for a listed company to disclose its ownership structure at the aggregate level. Data on both direct and indirect equity ownership of controlling owners is part of this disclosure under the heading “promoter holdings.” Under Clause 35, the total promoter holdings are subdivided into “shares owned by promoters” and “shares owned by persons acting in concert” (PACs). Although the term “promoter” is mentioned in Indian company law, it is not legally defined under the Companies Act, 1956 other than referring to persons/entities that assume the initial task of setting up a company. However, a promoter is clearly defined under the Disclosure and Investor Protection (DIP) Guidelines (2000) of SEBI from a “control” perspective,¹⁶ whereby the term promoter includes the person or persons who are in over-all control of the company, the person or persons who are instrumental in the floatation of a company. The term “control” is defined under Clause (c) of sub-regulation (1) of Regulation (2) of the Substantial Acquisition of Shares and Takeover Regulations 1997 (Takeover Code) and includes the “right to appoint the majority of the directors or to control the management or policy decisions exercisable by a person or persons acting individually or in concert, directly or indirectly, including by virtue of their shareholding or management rights or shareholding agreements or voting agreements or in any other manner.” During the period of our study, the direct holdings of promoters had to be disclosed under Clause 35 under the heading “Promoter” that was classified further into “Indian Promoters” and “Foreign Promoters.”

With regard to the indirect holdings of promoters, these were covered under the heading “persons acting in concert,” (PAC) as per Regulation 2(1) (e) (i) of the Takeover Code. As per the regulation, PACs are

¹⁶ The meaning of promoter and promoter group is defined as per Explanations I, II and III to sub-clause (m) of clause 6.8.3.2 of the DIP Guidelines 2000.

those, who for a common objective or purpose, engage in substantial acquisition of shares or voting rights or gaining control over a target company. For disclosure purposes as required by Clause 35, PACs essentially include holdings by individuals/entities that have a controlling element that adds to the control exercised by the promoters.¹⁷ Thus, PROM_CONTROL, the equity holding of controlling owners is defined as the sum total of promoter share and PACs. This data is sourced at the company level from the Prowess database for the years 2001 – 2006.

With regard to our second variable of interest, group affiliation, the Prowess database identifies firms by their association with a particular ownership group.¹⁸ Group affiliation is represented by the dummy variable GROUP defined at the company level in the following way: GROUP =1 if a firm is affiliated to a business group; GROUP = 0 otherwise. For all standalone or unaffiliated firms, therefore GROUP = 0.

3.3.2 Ownership Opacity

Based on Sarkar and Sarkar (2008), we construct two indices of ownership opacity. We exploit the information disclosed on individual equity holdings of the promoters to compute these indices. In addition to the disclosure of the aggregate ownership structure, Clause 35 of the Listing Agreement requires all listed companies to mandatorily disclose the identity of every equity holder who has *one percent or more* share ownership in the company along with the value of such ownership and the number of shares held. Thus, any discrepancy between the promoters' shareholding disclosed at the aggregate level and that summed up from the one percent disclosure is accounted by those promoter-shareholders with less than one percent ownership. It is with respect to these shareholders that we have no information on their identity or pattern of equity ownership. The more such divergence, the less we are likely to know the control and ownership structure of the controlling shareholders of the firm, and thus more opaque is the firm. To capture such opacity, which we term as Type 1 opacity, we construct the variable OPAQUE_1 as follows:

¹⁷ Specifically, as detailed by the Secondary Market Advisory Committee (SMAC, 2004) of the SEBI, PACs refers to entities who are in control of the company, directly or indirectly, whether as a shareholder (including shareholding of 10% or more in such companies), director or otherwise, or person or persons named as promoters in any document filed with SEBI, stock exchange, Registrar of Companies or any other body of offer of securities to the public or existing shareholders. Thus, PACs can include a company, its holding company or subsidiary company or such company under the same management either individually or together with each other, its subsidiaries and also the directors of such companies. <http://www.sebi.gov.in/commreport/clause35.html>

¹⁸ Prowess associates a company to a group based on company prospectus/letter of offer, annual reports, disclosure of related party transactions, company website, house journals, equity shareholding details of listed companies and the list published by the Monopolies and Restrictive Trade Practices (MRTP) Commission.

OPAQUE_1 = Total promoters' share ownership from aggregate data minus total promoters' share ownership from 1 percent data, divided by total promoters' share ownership from aggregate data.

Table 3 presents illustrates the construction of the OPAQUE_1 measure based on disclosed information of one percent equity holdings for two flagship companies belonging to top two business groups in India, namely Reliance Industries of the Reliance Group, and Tata Steel of the Tata Group. While aggregate promoter ownership in Reliance Industries is 46.76 percent, around half of this ownership is accounted for by promoters with less than one percent holdings and hence their identities and shareholding are not available in the public domain. In contrast, in the case of Tata Steel, more than 90 percent of the aggregate promoter shareholding is accounted for in the one percent disclosure records so that OPAQUE_1 for Tata Steel is much less, at 0.08 as compared to 0.39 for Reliance Industries.

The second indicator of opacity is derived from the structure of information on promoter ownership that is available in the one percent disclosures. Table 2 shows that there is considerable higher fragmentation of the ownership structure with regard to promoters and persons acting in concert (PACs) in the case of Reliance Industries compared to Tata Steel. While the ownership structure in Reliance Industries is more fragmented with one promoter and fourteen PACs, the ownership of Tata Steel is more consolidated with no PACs and two promoters. While fragmentation of ownership structure per se need not necessarily pose a problem in deciphering the inter linkages between group affiliates, the problem arises when many of the promoters and PACs are private unlisted companies so that the complete chain of control cannot be established. This is the case largely with Reliance, where five of seven and *all* fifteen promoters and PACs taken together are private limited companies respectively. In the case of Tata Steel, however, both promoters are listed companies.

To capture the fragmentation of one percent or more equity holdings by insiders, which we term as Type 2 opacity, we construct the measure OPAQUE_2:

OPAQUE_2 = (Total number of separate holdings by promoters of 1 percent or more) minus (total number of separate holdings by promoters of 5 percent or more) divided by total number of separate holdings by promoters of 1 percent or more.

The larger the magnitude of OPAQUE_2, the more is the fraction of ownership held by the relatively smaller shareholders with equity ownership of less than 5 percent. As in the case of OPAQUE_1, estimate of OPAQUE_2 is higher for Reliance Industries as compared to Tata Steel. It is important to note in this

context, that high values of OPAQUE_1 need not necessarily imply high values of OPAQUE_2. In this sense, each of these two measures carries independent information.

Insert Table 3 about here

3.4 Control Variables

Opportunistic earnings management may be affected by factors other than our main variables of interest. To control for such effects we use the well known firm level attributes that have been documented in prior literature (Warfield et al., 1995; Ali et al., 2007). These include leverage (LEVERAGE), price to book ratio (PB), firm size (SIZE), systematic risk of equity (BETA), cash flow from operations (CFO), lagged total current accruals (L1ACCR), return on assets of current year (ROA) and past years (PROA), variance in quarterly earnings (VAREPS), and the absolute level of return on assets (ABS_ROA) as well and its growth (ABS_ROA_GR). In addition, we consider two ownership measures relevant to the emerging markets namely the percentage of equity ownership by foreign institutional investors (FII_SHR) and banks and financial institutions (BANKS_FI_SHR). Finally, given that our period of analysis coincides with the inception and progression of governance reforms under Clause 49, we control for this effect by incorporating a dummy variable POST_CG49 which takes the value 1 from the year the particular firm adopted the Clause 49 regulations. All regressions include industry dummies and year dummies to control for unobserved industry and time specific effects. Details of variables description are given in Table 4.

Insert Table 4 about here

4. Empirical Models

This section presents the earnings management models that we use to study the effects of insider control, group affiliation and ownership opacity on discretionary accruals.

4.1 Benchmark Model of Opportunistic Earnings Management

Our benchmark model of discretionary accruals is:

$$\begin{aligned}
 DA = & \alpha + \beta_1 FII_SHR + \beta_2 BANKS_FI_SHR + \beta_3 LEVERAGE + \beta_4 PB + \beta_5 SIZE + \beta_6 BETA \\
 & + \beta_7 CFO + \beta_8 L1ACCR + \beta_9 ROA + \beta_{10} PROA + \beta_{11} VAREPS + \beta_{12} ABS_ROA + \beta_{13} \\
 & ABS_ROA_GR + \beta_{14} POST_CG49 + \sum_i \gamma_i INDUSTRY_i + \sum_t \delta_t YEAR_t + error \quad (M1)
 \end{aligned}$$

where, DA is the amount of discretionary accruals estimated using the Francis et al. model and the explanatory variables are as outlined in the previous section. We examine both unsigned as well as signed discretionary. Further, we examine two subsamples with only positive and only negative discretionary accruals. There is a benefit to examining both the unsigned and the signed measures since they have different implications. While unsigned DA can be interpreted to be representing income smoothing, signed DA represents the earnings overstatement efforts by the firm. The motivations for income smoothing and earnings overstatement are different and need to be distinguished for the scenarios we are trying to analyze. For a standalone firm trying to meet investor expectations of consistent financial performance, income smoothing may be an appropriate criterion to examine. However, for a group-affiliated firm trying to provide funds to another firm, overstating income may be more appropriate. Since we consider both non-group (i.e. stand alone) firms with promoter influence as well as group firms in our analysis, there is a role for both measures to be examined, based on the issue we are focusing on.¹⁹

4.2 Effect of Insider Control on Earnings Characteristics

To identify the relative strengths of the alignment and entrenchment effect of promoter control on discretionary accruals we incorporate a quadratic term for the PROM_CONTROL variable along with other control variables as follows:

$$\begin{aligned}
 DA = & \alpha + \phi_1 PROM_CONTROL + \phi_2 PROM_CONTROL^2 + \beta_1 FII_SHR + \beta_2 BANKS_FI_SHR \\
 & + \beta_3 LEVERAGE + \beta_4 PB + \beta_5 SIZE + \beta_6 BETA + \beta_7 CFO + \beta_8 LICACCR + \beta_9 ROA + \\
 & \beta_{10} PROA + \beta_{11} VAREPS + \beta_{12} ABS_ROA + \beta_{13} ABS_ROA_GR + \beta_{14} POST_CG49 + \sum_i \gamma_i \\
 & INDUSTRY_i + \sum_t \delta_t YEAR_t + error
 \end{aligned}
 \tag{M2}$$

If $\phi_1 < 0$ and $\phi_2 > 0$, the relationship between ownership and discretionary accruals is U-shaped. This implies that earnings management reduces with increases in ownership, when ownership is below a certain threshold level, and increasing thereafter. The alignment effect outweighs entrenchment effect at lower levels of ownership; the opposite holds true when controlling shareholders consolidate their

¹⁹ Frankel et al. (2002) suggest that income-increasing (positive) discretionary accruals are likely to be more opportunistic and that auditors are more likely to require adjustments to positive rather than negative accruals. However, Bedard et al. (2004) argue that aggressive earnings management includes both positive and negative discretionary accruals. The existence of bonus caps can make managers move current profits to the future by creating “cookie” jars. Further, managers can take “big baths” to generate negative discretionary accruals in periods of negative earnings so that the future turnaround looks impressive. Several papers including Klein (2002), Haw et al. (2004) have examined similar issues using absolute measures of discretionary accruals. We consider both signed and unsigned as opportunistic, interpreting the former as overstatement and the latter as smoothing.

ownership beyond that threshold. If $\phi_1 > 0$ and $\phi_2 < 0$ then the entrenchment effect is stronger than the alignment effect at lower levels of ownership and becomes weaker at higher levels of ownership.

4.3 Effect of Business Group Affiliation on Earnings Characteristics

We use a binary variable, GROUP, to capture the effect of group affiliation. To examine the group effect, we first estimate model (M2) above by dropping the promoter control variable *PROM_CONTROL* and incorporating GROUP as in the following specification (3).

$$DA = \alpha + \phi_1 GROUP + \beta_1 FII_SHR + \beta_2 BANKS_FI_SHR + \beta_3 LEVERAGE + \beta_4 PB + \beta_5 SIZE + \beta_6 BETA + \beta_7 CFO + \beta_8 LICACCR + \beta_9 ROA + \beta_{10} PROA + \beta_{11} VAREPS + \beta_{12} ABS_ROA + \beta_{13} ABS_ROA_GR + \sum_i \gamma_i INDUSTRY_i + \sum_t \delta_t YEAR_t + error \quad (M3)$$

We then incorporate back the quadratic term of *PROM_CONTROL* as in (2) so that the full model for estimating the effect of insider control on DA is as follows:

$$DA = \alpha + \phi_1 PROM_CONTROL + \phi_2 PROM_CONTROL^2 + \phi_3 GROUP + \beta_1 FII_SHR + \beta_2 BANKS_FI_SHR + \beta_3 LEVERAGE + \beta_4 PB + \beta_5 SIZE + \beta_6 BETA + \beta_7 CFO + \beta_8 LICACCR + \beta_9 ROA + \beta_{10} PROA + \beta_{11} VAREPS + \beta_{12} ABS_ROA + \beta_{13} ABS_ROA_GR + \sum_i \gamma_i INDUSTRY_i + \sum_t \delta_t YEAR_t + error \quad (M4)$$

We further investigate the effect of group affiliation by splitting the sample into group and non-group firms and estimate model (M1) separately for these two subgroups to examine whether the effect of insider control on opportunistic earnings management is fundamentally different between these subgroups.

4.3 Effect of Ownership Opacity on Discretionary Accruals

Finally, we introduce the measures of ownership opacity, OPAQUE_1 and OPAQUE_2 in our benchmark DA model given by specification (M1) and estimate the effect of each measure on discretionary accruals after controlling for group effects. Thus we estimate the following model:

$$DA = \alpha + \phi_1 OPAQUE_i + \phi_2 GROUP + \beta_1 FII_SHR + \beta_2 BANKS_FI_SHR + \beta_3 LEVERAGE + \beta_4 PB + \beta_5 SIZE + \beta_6 BETA + \beta_7 CFO + \beta_8 LICACCR + \beta_9 ROA + \beta_{10} PROA + \beta_{11} VAREPS + \beta_{12} ABS_ROA + \beta_{13} ABS_ROA_GR + \sum_i \gamma_i INDUSTRY_i + \sum_t \delta_t YEAR_t + error \quad (M5)$$

5. Empirical Results: Effect of Insider Control and Group Affiliation

5.1 Descriptive Statistics

We present two sets of descriptive statistics for our sample in Tables 5.1 to 5.2. In Table 5.1, we present means for all the variables used in our analysis. The means are presented for the sample as a whole and separately for group affiliated and standalone firms. In Table 5.2, we examine the pervasiveness of insider control for our sample of firms in terms of different control thresholds and how blockholding by insiders have changed over the period of our study.

--- Insert Table 5.1 here ---

Considering the entire sample, Table 5.1 shows that absolute discretionary accruals is around 8 percent of total assets. The positive and negative discretionary accruals are of a similar magnitude in the range of 7 percent – 9 percent of total assets, thus yielding almost close to zero signed discretionary accruals. The mean promoter ownership is about 50 percent. Compared to that, foreign institutional investors (FII_SHR) have only 1.6 percent ownership and banks and financial institutions (BANKS_FI_SHR) own only about 3.7 percent of total equity. Focusing on control variables, mean firm size (log of market capitalization) (SIZE) is 18.8 and the mean leverage (LEVERAGE) is 0.32. The 12 quarter variation in earnings per share (VAREPS) is quite low, but there is considerable variation. The price-to-book ratio (PB) is around 1. Cash flow from operations (CFO_TA) is about 6% of total assets. The return on assets (ROA) is 13% and the 5-year return on assets (PROA) is 2.3%.

Examining differences in characteristics by ownership groups, there are no statistically significant differences between group and non-group firms in both absolute and signed discretionary accruals. In case of promoter control (PROM_CONTROL) group affiliates exhibit higher control at 50 percent as compared to 48 percent for standalones, with the difference being significant at the 5 percent level. Further, foreign institutional investors (FII_SHR) and banks and financial institutions (BANKS_FI_SHR) have higher equity ownership in group affiliates than standalones with the differences being significant at 1 percent. Group affiliates are also larger in size (SIZE) than standalones, have higher profitability (ROA), higher market valuation (PB), higher cash flow (CFO_TA) and are more leveraged (LEVERAGE).

Insert Table 5.2 about here

Concentrated insider ownership is predominant in Indian firms. Table 5.2 presents the distribution of firms by different cut-offs of insider control for each of the five years included in our study. At the 5 percent control threshold, one of the most common cut-off points used in the literature to define blockholdings²⁰, 98 percent of firm year observations have insiders holding five percent or more. This percentage has remained almost the same for sample firms in each of the years covered by the study. Increasing the cut-off to 20 percent which is typically considered as the holding where a block of shareholders can get effective control, here too, one finds around 94 percent of sample observations have effective insider control. About 90 percent of sample firms on an average have insider control at least equal to 26 percent required to block special resolutions²¹ and around 47 percent of sample firms have majority control. The estimates in Table 4.2 thus shows that insider control is pervasive in Indian firms; corresponding estimates obtained for group affiliates and standalones (not reported in the Table) paint a similar picture.

5.2 Regression Results

5.2.1 Benchmark model

Table 6.1 presents the regression results for the benchmark model. Bank ownership reduces absolute discretionary accruals but increases signed discretionary accruals. This is because higher bank ownership leads to less negative discretionary accruals. Higher leverage leads to increase in absolute discretionary accruals, but to lower values of signed as well as negative discretionary accruals. In essence, higher leverage makes firms report more smooth and conservative earnings. Higher operating cash flows also lead to lower signed discretionary accruals, whereas higher current accruals from a year ago lead to higher signed discretionary accruals. Higher return on assets leads to lower absolute discretionary accruals but higher discretionary accruals. Absolute return on assets is positively (negatively) correlated with the unsigned (signed) discretionary accruals. Overall, the estimates are consistent with those found in earlier studies and confirm the validity of the benchmark model in the Indian setting.

--- Insert Table 6.1 here ---

²⁰ This is drawn from the legal definition of blockholders under Rule 13d-1(a) of the Securities Exchange Act of 1934 in the US.

²¹ Under the Companies Act of 1956, important decisions like changing the line of business, diversification, amalgamation, alteration of shareholder rights, reduction in share capital, etc. have to be approved by a special resolution which is deemed to be accepted if “the votes cast in favor of the resolution ... are not less three times the number of the votes, if any, cast against the resolution by the members so entitled and voting, (Section 189).”

5.2.2 Effect of insider control

The effect of insider control in terms of the alignment and entrenchment effects are presented in Table 6.2. Recall that the relative strength of these two effects are captured by the quadratic specification of the promoter control variables namely, $PROM_CONTROL$ and $PROM_CONTROL^2$. The coefficient estimates presented in Table 5.2 show that insider control has non-linear effect on income smoothing, as measured by absolute discretionary accruals. Given that the coefficient of $PROM_CONTROL$ is negative and that of $PROM_CONTROL^2$ is positive, both being statistically significant at 1 percent, an increase in insider control reduces income smoothing with the relationship reaching a minimum when promoter control is around 50 percent, and then starts increasing thereafter. This result suggests that the alignment effect outweighs the adverse effect until insider control reaches an absolute majority, and reversing thereafter. This U-shaped relationship of insider control is however not obtained with respect to signed discretionary accruals, irrespective of whether we consider the positive and negative values separately or together. The influence of the control variables on discretionary accruals continue to hold as in the benchmark model.

Insert Table 6.2 about here

The U-shaped relationship that we obtain between insider control and absolute discretionary accruals is consistent with the result obtained by Wang (2006) with respect to a sample of large US family firms. However, there are two important qualitative differences. First, the study by Wang finds evidence of both a statistically significant linear and positive relationship between family control and absolute discretionary accruals as well as evidence of a U-shaped relationship with a turning point of around 34 percent. These two apparently incongruous findings can be reconciled by noting that the mean (median) family equity ownership in the study (Wang, 2006) is 10.5 (5.0) percent so that most of the sample observations are less than the threshold value of 33 percent. Thus, the non-linear relationship is driven by a few firms at the higher end of the distribution. In our case, the finding of the non-linear relationship is unique in the sense that a linear specification leads to insignificant coefficient of the promoter control variable. Second, the threshold value (or the turning point) in our study is around 50 percent which almost coincides with both the mean and median values of equity ownership. This suggests that the entrenchment effect outweighs the alignment effect for nearly half of the sample observations which occurs when insiders have near absolute control over firm operations.²²

²² The average equity holding for observations with at least 50 per cent insider control is around 51 per cent and the estimate jumps to around 64.5 per cent for sample observations with insider control of at least 51 per cent (Table 4.3).

Our result of a U-shaped relationship between insider control and absolute DA is in contrast to the inverted U relationship obtained with respect to a sample of listed private sector firms in China (Ding et al., 2007). In that study, the interests of controlling insiders converge with those of outsiders when insider control exceeds absolute majority (between 55 and 60 percent). However, in China most listed companies have significant government ownership and business group companies are hardly dominant. In China, most companies are organized as a parent with a single subsidiary company. The subsidiary company is usually listed to provide access to external finance for expansion of the parent company (Fan, 2008). Accordingly, pyramidal ownership structures with complex cross holdings manifesting in significant divergence between ownership and control are less prevalent in Chinese companies. The contrasting results obtained in our study highlights the importance of organizational forms in influencing agency costs and accordingly observed economic outcomes.

5.2.3 Effect of Group Affiliation on Earnings Characteristics

The effect of business group affiliation on discretionary accruals is estimated in terms of Model 3 and Model 4 set out in Section 4. In Model 3, we capture the effect of group affiliation using a binary variable, GROUP and test the effect of group affiliation to the exclusion of promoter control. Table 6.3 presents these regression results.

Group affiliation has a positive effect on absolute discretionary accruals with the coefficient being statistically significant at 5 percent. In other words, relative to standalone firms, group firms engage in more income smoothing practices. This result reinforces the findings in the literature that group affiliation adversely affects the quality of earnings due to opportunistic earnings management by insiders (Kim and Yi, 2005; Siregar and Utama, 2008). Group affiliation however, does not seem to affect the other measures of discretionary accruals – unsigned, positive or negative.

Insert Table 6.3 about here

Table 6.3 provides preliminary evidence on the differential behavior of group firms in earnings management. However, this result depends on assuming that the coefficients of all the other parameters are same across group and non-group firms. To further examine the robustness of the above finding, we split our sample by group firms and non-group firms and estimate the effect of insider control for each sub-sample. The results of this estimation are presented in Tables 7.1 and 7.2, respectively. Table 7.1 shows that the coefficient of PROM_CONTROL is negative and that of PROM_CONTROL² is positive, with both coefficients statistically significant, at 5 percent and 1 percent, respectively. The magnitude of

the coefficients suggests that the threshold value (or turning point) is at 53.24 percent which is close to the level that we find in the full sample. Compared to these findings for group-affiliated firms, the estimates in Table 7.2 show that though the coefficients of PROM_CONTROL is negative and that of PROM_CONTROL² is positive for standalone firms, these coefficients are significant only at the 10 percent level. Thus, our separate estimations of group and standalone firms suggest that the U-shaped relationship that we find in our full sample is driven mostly by the group affiliated firms. Put differently, there seems to be a strong group effect in the relation between discretionary accruals and group affiliation.

Insert Tables 7.1 and 7.2 about here

The final exercise that we conduct to get additional insights into the effect of group affiliation is by splitting the sample observations by different thresholds of insider control. Insider controls in group firms can be exercised at relatively low levels of promoter ownership because of pyramiding. Table 6.3 points to an adverse group effect for the entire sample, but it does not tell us at level of insider control does the adverse group affiliation sets in. For this, we first split the sample by three different control thresholds namely, 20 percent, 26 percent, 51 percent. Corresponding to each control threshold, we define LOW to be the set of firms with promoter ownership less than the threshold and HIGH to be the set of firms with equity ownership equal or above that threshold. We then re estimate Model 3 with the GROUP variable for the LOW and HIGH sub-samples for each of the three thresholds identified above. Table 8 presents the results of these six regressions. As our main interest is on the effect of group affiliation, we present the parameter estimates of only the GROUP variable.

Insert Table 8 about here

For both 20 percent and 26 percent threshold levels, group affiliation has no effect in firms with promoter control below the thresholds (i.e., for LOW firms). In contrast, the coefficient of GROUP is positive and significant in the HIGH sample suggesting that group affiliation increases discretionary accruals in firms when promoter control exceeds the thresholds of 20 percent and 26 percent. For the 51 percent threshold, the coefficient on the GROUP variable is positive and significant for both the LOW and the HIGH sub-samples. Very interestingly, the coefficient on the GROUP variable in the HIGH sample is about twice the value of the corresponding estimates with the 20 percent and 26 percent thresholds. These results strongly suggest that group affiliation imposes additional costs in terms of earnings management when promoter control exceeds some minimum thresholds that give them effective control over the firm.

5.2.5 Robustness checks

We tested if our results are sensitive to the model used to measure discretionary accruals. Instead of using the Francis et al. (2005) model, we used the Jones (1991) model to measure discretionary accruals and found our conclusions to be unchanged. We also examined if our results were sensitive to the definition of firm size we have used, by re-estimating the regressions using market capitalization of the firm as a measure of size. Again, all our results remain robust.

6. Effect of Ownership Opacity: Descriptive Statistics and Regression Results

Our analysis of ownership opacity and opportunistic earnings management is based on the sample of listed firms for which promoter ownership of one percent or more is available. The analysis covers the period from 2002-03, the first year the disaggregated data is available, to 2004-05, the latest year for which data was available at the time of analysis. As outlined earlier, we consider two measures of ownership opacity, OPAQUE_1 and OPAQUE_2. While OPAQUE_1 seeks to capture opaqueness of insider control due to partial disclosure of promoter holdings, OPAQUE_2 captures the opaqueness that arises from fragmentation of promoter holdings.

--- Insert Table 9.1 about here ---

6.1 Descriptive Statistics

Table 9.1 presents the summary statistics of the sample used for the opacity analysis both in the aggregate as well as according to ownership groups. For the full sample, the mean value of OPAQUE_1 is 0.2584 and that of OPAQUE_3 is 0.1860. Comparing the opacity measures by ownership groups, one finds that the mean opacity by both measures to be higher for standalones as compared to group affiliates, with the difference in each case being statistically significant at the one percent level. Differences between ownership groups with respect to other variables, including absolute discretionary accruals, are more or less in line with those in the larger sample (Table 5.1).

6.2 Regression Results

Tables 9.2 and 9.3 present the regression results with respect to ownership opacity and absolute discretionary accruals estimated using Model 5. Table 9.2 shows coefficient estimates with regard to OPAQUE_1 and OPAQUE_2 as well as other control variables for the full sample of 2736 firm year observations in the case of OPAQUE_1, and 2386 firm year observations in the case of OPAQUE_2. The first two columns of Table 9.2 present coefficient estimates without considering the effect of

promoter control. The third and fourth column present coefficient estimates after specifying a quadratic relationship with PROM_CONTROL as in our original estimation.

As is evident from Columns 1 and 2 of the Table, neither ownership opacity measures have a statistically significant effect on absolute discretionary accruals. In other words, the non-disclosure of promoter equity below one percent and the fragmentation of promoter ownership do not seem to have any impact on opportunistic behavior of controlling insiders. Similar results hold with respect to the opacity measures after taking into account the effect of insider control (Columns 3 and 4). However, with regard to the effect of PROM_CONTROL, while the U-shaped relationship is preserved when we consider OPAQUE_1, only a linear negative relationship with respect to absolute discretionary accruals is obtained when we consider OPAQUE_2. Thus, when we consider the full sample, i.e. pool group affiliates and standalone firms, we do not find any relation between opacity on discretionary accruals.

--- Insert Tables 9.2 and 9.3 about here ---

However, while opacity can impact both standalone and group affiliates as the concept is essentially related to the extent of disclosure of information of promoter ownership stakes, it is our hypothesis that the complex group structure associated with group affiliates via pyramiding and cross-holdings can compound the effect of non-disclosure and fragmentation of insider control in group firms as compared to standalones. This despite the fact that as revealed by summary statistics, the average opacity levels are significantly higher in standalones as compared to group affiliates. However, once characteristics of group firms and standalones are controlled for in a multivariate framework, it is our expectation that the marginal effect of opacity on earnings management DA would be higher for group affiliates as compared to that of standalones.

To test this hypothesis, we now estimate the effect of OPAQUE_1 and OPAQUE_2 separately for group affiliated and standalone firms. The results of this estimation are presented in Table 9.3. We observe that OPAQUE_1 now has a positive and statistically significant relation with absolute discretionary accruals for group affiliated firms, while there is no such relation for standalones. However, when we consider OPAQUE_2, it is neither significant in the case of group affiliated or standalone firms. In other words, the type of ownership opacity, as well as the type of ownership matters for opportunistic earnings management. It is worth noting that the estimations undertaken in Table 9.3 consider ownership opacity along with indicators of insider control. If we drop the two variables with promoter control (the linear and the quadratic terms) then the coefficients attached to both OPAQUE_1 and OPAQUE_2 become

significant for group affiliated firms while they continue to be insignificant for the standalone firms. Turning to how insider control impacts absolute discretionary accruals in ownership groups in the presence of opacity, we find from Table 9.3 that the U-shaped relationship is preserved only in the case of group affiliates irrespective of which measure of opacity we consider. In contrast, insider control ceases to have any effect on absolute discretionary accruals in standalones once we explicitly consider opacity. Taken together, these results suggests that ownership opacity which captures the structure of ownership is a significant determinant of absolute discretionary accruals in group affiliated firms, over and above the effect that is exerted through the level of promoter control.

Given that opaqueness of insider control has statistically significant effect on opportunistic earnings management in group affiliated firms, with the U-shaped relationship with insider control preserved, we investigate the issue of whether the level of insider control in group affiliates has any effect on how opaqueness impacts earnings management. To examine this, we take the sample of group affiliated firms for the three years under consideration and split the sample by insider control thresholds of 20 percent, 26 percent and 51 percent. For each split, we estimate the effect of opaqueness on absolute discretionary accruals. Tables 10.1 and Table 10.2 present the estimation results with respect to OPAQUE_1 and OPAQUE_2, respectively.

--- Insert Tables 10.1 and 10.2 about here ---

An examination of the coefficient estimates for OPAQUE_1 for each threshold interestingly reveals that OPAQUE_1 has an effect on absolute discretionary accruals only for firms where insider control less than 51 percent. Combining this with the finding that OPAQUE_1 has no effect for firms when promoter control exceeds the thresholds of 26 percent or and 51 percent implies, that it is between the range of greater than 26 percent and less than 51 percent that OPAQUE_1 increases absolute discretionary accruals. Thus, for the range that insiders have relatively low equity ownership (< 26 percent), incomplete disclosure of promoter ownership has no bearing on opportunistic earnings management. One can interpret this in terms of insiders having insufficient control to manage earnings by leveraging ownership opacity. This is also the range where the alignment effect is strong enough to outweigh any negative entrenchment effect. On the other hand, for the range when insiders can leverage on absolute control (≥ 51 percent) to potentially expropriate minority investors through opportunistic earnings management, and entrenchment effect is found to be strong enough to outweigh the benefits from convergence of interests, our results that non-disclosure of ownership information has little marginal effect on absolute discretionary accruals makes intuitive sense. For the intermediary range of 26 to 51, when the positive

effects of alignment of interests are tapering off and stronger entrenchment effects are setting in, the ultimate owners with less than majority cash flow rights in pyramidal set ups can successfully extract private benefits through tunneling by concealing the ownership structure.

Turning to the effect of OPAQUE_2 on absolute discretionary accruals (Table 10.2), we find a different set of results as compared to that with respect to OPAQUE_1. When we consider the effect at thresholds of less than 20 percent and 26 percent, like OPAQUE_1, we find no effect of OPAQUE_2 on absolute discretionary accruals for low levels of insider control. However, we observe that for the 51 percent threshold, while OPAQUE_2 has no effect for ownership stakes below 51 percent, the coefficient of OPAQUE_2 is highly significant once promoters acquire majority stakes. Recall that OPAQUE_2 measure captures ownership fragmentation *within* the declared ownership structure. When promoters hold significantly high (declared) stakes in a company, one way of creating non-transparent ownership structure is through complex webs of control through cross-holdings and pyramiding which can be achieved by a large number of entities with each having small shareholdings. The fact that many of these smaller entities could be private limited companies which make it difficult to follow the ownership trail (as private companies are not required to disclose ownership patterns) makes this fragmentation an effective tool to conceal the true ownership information. Our result suggests that such complexity is strongly correlated with indicators of earnings management in group affiliated companies.

6.3 Robustness checks

As with the earlier approach, we test if our results with respect to opacity are sensitive to the model used to measure discretionary accruals. Again, instead of using the Francis et al. (2005) model, we used the Jones (1991) model to measure discretionary accruals and found our conclusions to be unchanged. Additionally, we examined if our results were sensitive to the definition of firm size we have used, by re-estimating the regressions using market capitalization of the firm as an alternative measure of size. Again, all our results remain robust.

7. Conclusion

Our empirical analysis of insider control and opportunistic earnings management in emerging economies like India has sought to add to the sparse evidence that exists on this issue with respect to these countries where insider control in the corporate sector is a dominant feature. We studied three important aspects of insider dominance, namely promoter control, the effect of group affiliation and the impact of opacity on discretionary accruals.

We find consistent and robust evidence of a non-linear U-shaped relationship between promoter control and discretionary accruals which suggests that the relative strengths of the alignment effect vis-à-vis the entrenchment effect changes with changes in the levels of insider control with the entrenchment effect dominating the alignment effect once controlling owners acquire absolute (majority) control of the firms. This relationship is largely driven by the group-affiliated firms. Our results of a U-shaped relationship are qualitatively consistent with the findings with respect to large founding family firms (Wang, 2006) but negate the findings with respect to Chinese private sector listed firms which we ascribe to the difference the organization forms of corporate ownership in these two countries.

The second set of empirical analysis in our paper focuses on the effect of group affiliation on discretionary accruals. Consistent with existing evidence on emerging economies, we find that group affiliation increases earnings opportunism of insiders. We gain additional insights into the effect of group affiliation through our threshold analysis. The results reveal that the adverse effect of group affiliation sets in only when insider control reaches the critical minimum. Given that one of the channels through which group firms engage in earnings management is through internal capital markets, we also provide some supporting evidence on this account by showing that higher intra-group financing activity in group affiliated firms is strongly correlated with higher levels of discretionary accruals.

Finally, we examine how ownership opacity which is an important measure of the complexity of insider control, affects opportunistic earnings management. Our results show that opacity of insider ownership manifested in fragmentation of equity amongst a large number of inside owners increases the discretionary accruals. Again this effect is driven by the behavior of the group affiliated firms with the effect getting amplified with higher levels of control thresholds. In conclusion, our results highlight that both promoter control and group affiliation may independently influence agency costs in emerging economies. This in turn calls for policy actions that focus not only on individual firms but business group firms as consolidated identities.

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Table 1: Ownership and Control Characteristics, Earnings Management, Shareholder Protection and Select Institutional Indicators: India and the US

	India	US
I. Percentage of widely held firms (top 20 listed firms) ¹	10	80
II. Percentage of family controlled firms (top 20 listed firms)	90	20
III. Mean ownership concentration ²	43	12
IV. Characteristics of Family Business Groups ³		
(i) Percentage of group affiliated firms controlled through pyramids	10.02	0.90
(ii) Percentage of market capitalization held by group controlled firms	20.89	3.04
(iii) Percentage of market capitalization held by pyramid controlled firms	4.10	0.69
(iv) Pyramid Layer	0.36	0.31
V. Aggregate Earnings Management Score ⁴	19.1	2.0
VI. Shareholder Protection ⁵		
(i) Oppressed minority mechanism	1	1
(ii) Antidirector Rights of Minority Shareholders ^a (scale of 6)	5	5
VII. Institutional Indicators ⁶		
(i) Efficiency of Judicial System	8	10
(ii) Rule of Law	4.17	10
(iii) Auditor's per 100,000	9	168
(iv) Disclosure Level	61	85
(v) Opacity in Accounting	79	25
(vi) Accounting Standard	57	71

Source: 1: The data for India is computed from Prowess and the data for US is sourced from La Porta et al. (1999). 2: Leuz et al. (2003); 3: Masulis et al., (2009); 4: Bhattacharya et al. (2003); 5: La Porta et al. (1999); 6: La Porta, et al. (1998); (2000).

Table 2: Sample Selection Steps

	Firm year	Firms
Step 1: All BSE listed firms with the following criteria (i) Non-agricultural industries (ii) Basic accounting and market data available for 2001-06 (iii) Not government or foreign listed (iv) No outliers for leverage, ROA and abs growth in ROA (v) Not Z listed in the BSE	10749	2431
Less: missing ownership data	909	

Step 2	9840	
Less: missing data for DA, ERC and control variables	3952	

Step 3: Final sample	5888	1658

Note: Outliers are those that have either (i) leverage < 0 or leverage > 10, (ii) roa > 100% or roa < -100%, (iii) abs_roa_gr > 1000% or abs_roa_gr < -1000%

Table 3: Computation of Opacity Measures (Year 2004)

Name of Affiliate	Type of Insider	Identity of Insider	Percentage Share
Reliance Industries Ltd.	Indian Promoters	Petroleum Trust	7.51
Reliance Industries Ltd.	Persons Acting In Concert	Sanchayita Mercantile Pvt Ltd	2.46
Reliance Industries Ltd.	Persons Acting In Concert	Reliance Enterprises Pvt Ltd	2.26
Reliance Industries Ltd.	Persons Acting In Concert	Florentine Trading Pvt Ltd	1.87
Reliance Industries Ltd.	Persons Acting In Concert	Velocity Trading Pvt Ltd	1.77
Reliance Industries Ltd.	Persons Acting In Concert	Madhuban Merchandise Pvt Ltd	1.75
Reliance Industries Ltd.	Persons Acting In Concert	Ornate Traders Pvt Ltd	1.4
Reliance Industries Ltd.	Persons Acting In Concert	Reliance Polyolefins Pvt Ltd	1.37
Reliance Industries Ltd.	Persons Acting In Concert	Tresta Trading Pvt Ltd	1.19
Reliance Industries Ltd.	Persons Acting In Concert	Reliance Capital Ltd	1.18
Reliance Industries Ltd.	Persons Acting In Concert	Amur Trading Pvt Ltd	1.18
Reliance Industries Ltd.	Persons Acting In Concert	Yangste Trading Pvt Ltd	1.16
Reliance Industries Ltd.	Persons Acting In Concert	Reliance Energy & Project Development Pvt Ltd	1.15
Reliance Industries Ltd.	Persons Acting In Concert	Reliance Aromatics & Petrochemicals Pvt Ltd	1.15
Reliance Industries Ltd.	Persons Acting In Concert	Reliance Chemicals Pvt Ltd	1.05
Reliance Industries Ltd	Total Insider Ownership : Aggregate Ownership Data (Z1)		46.76
	Total Insider Ownership : One Percent Data (Z2)		28.45
	OPAQUE_1 = (Z1– Z2)/Z1		0.39
	Total number of separate holdings by promoters of 1 percent or more (Y1)		15
	Total number of separate holdings by promoters of 5 percent or more (Y2)		1
	OPAQUE_2 = (Y1 – Y2)/Y1		0.93
Tata Steel	Indian Promoters	Tata Sons Ltd	19.8
Tata Steel	Indian Promoters	Tata Motors Ltd	4.66
Tata Steel	Total Insider Ownership : Aggregate Ownership Data (Z1)		26.56
	Total Insider Ownership : One Percent Data (Z2)		24.46
	OPAQUE_1 = (Z1– Z2)/Z1		0.08
	Total number of separate holdings by promoters of 1 percent or more (Y1)		2
	Total number of separate holdings by promoters of 5 percent or more (Y2)		1
	OPAQUE_2 = (Y1 – Y2)/Y1		0.50

Source: Computed from Prowess

Table 4: Variable Descriptions

Variable	
Discretionary accruals	
abs_DA_Francis	absolute value of discretionary accruals using the Francis et al. (2005) model
res_Francis_TA	signed value of discretionary accruals using the Francis et al. (2005) model
pos_DA_Francis	signed value of discretionary accruals using the Francis et al. (2005) model if it is >0; missing otherwise
neg_DA_Francis	signed value of discretionary accruals using the Francis et al. (2005) model if it is <0; missing otherwise
abs_DA_Jones	absolute value of discretionary accruals using the Jones (1991) model
res_Jones_TA	signed value of discretionary accruals using the Jones (1991) model
pos_DA_Jones	signed value of discretionary accruals using the Jones (1991) model if it is >0; missing otherwise
neg_DA_Jones	signed value of discretionary accruals using the Jones (1991) model if it is <0; missing otherwise
Ownership	
prom_control	percentage of equity ownership by promoters (direct promoter equity + equity of persons acting in concert)
fii_shr	percentage of equity ownership by foreign institutional investors
banks_fi_shr	percentage of equity ownership by banks and financial institutions
group	dummy variable; group =1 if firm is affiliated to business group, zero otherwise.
opaque_1	opaque_1= Total promoters' share ownership from aggregate data minus total promoters' share ownership from 1 percent data, divided by total promoters' share ownership from aggregate data.
opaque_2	opaque_2= Total number of separate holdings by promoters of 1 percent or more minus total number of separate holdings by promoters of 5 percent or more divided by total number of separate holdings by promoters of 1 percent or more.
Control Variables	
size	log of market capitalization
nrsize	percentile ranking of markets capitalization, scaled to range between -0.5 and +0.5
leverage	ratio of debt to total assets
vareps	variance of earnings per share over the prior 12 quarters
pb	market capitalization divided by book value of equity
bets	beta computed using the single factor CAPM
cfo_ta	operating cash flows divided by total assets from the previous year
L1caccr_ta	1 year lagged value of current accruals divided by total assets from the previous year
roa	return on assets
proa	average return on assets from the prior 5 years
abs_roa	absolute value of return on assets
abs_roa_gr	absolute value of growth in return on assets
postcg_49	corporate governance regulation dummy; equal to one if firm has adopted regulation, zero otherwise.

Table 5.1: Summary Statistics

Variable	All Firms		By Ownership group		
	N	Mean	Standalones	Group Affiliated	t-diff
abs_DA_Francis	5525	0.0835	0.0833	0.0837	0.16
res_Francis_TA	5525	0.002	0.0042	-0.0001	-1.24
pos_DA_Francis	2977	0.0794	0.0791	0.0796	0.15
neg_DA_Francis	2548	-0.0883	-0.0884	-0.0882	0.05
prom_control	5888	0.4911	0.48	0.5027	4.78
fii_shr	5888	0.0158	0.0061	0.0259	15.36
banks_fi_shr	5888	0.0367	0.0169	0.0574	25.46
size	5888	18.7956	17.9056	19.7245	38.84
nsize	5888	0.0446	-0.0874	0.1823	44.39
leverage	5888	0.3212	0.2834	0.3607	12.3
vareps	5888	0.4089	0.2058	0.6208	12.07
pb	5888	1.0676	0.911	1.2311	1.56
beta	5888	1.0169	0.9758	1.0598	5.2
cfo_ta	5783	0.0573	0.049	0.0658	4.65
l1caccr_ta	5778	0.0054	0.0115	-0.0007	-3.45
roa	5888	0.1302	0.1241	0.1365	3.74
proa	5888	0.0238	0.0225	0.0251	1.23
abs_roa	5888	0.1449	0.1429	0.1469	1.41
abs_roa_gr	5888	0.6996	0.776	0.6199	-5.26

Note: Number of standalone firm observations: 3007; Number of group-affiliate firm observations: 2881

Table 5.2: Distribution of Firm Year Observations by Insider Control Thresholds: 2001-06

Insider Control Thresholds (%)	Percentage of Observations by Year					Percentage of All Firm-Year Observations
	2002	2003	2004	2005	2006	2002-06
>= 5	99.85 (909)	98.45 (1083)	98.32 (1172)	98.23 (1227)	98.46 (1403)	98.40 (5794)
>= 20	95.01 (876)	94.54 (1040)	94.21 (1123)	93.75 (1171)	93.68 (1335)	94.17 (5545)
>= 26	88.39 (815)	89.90 (989)	90.43 (1078)	89.83 (1122)	89.61 (1277)	89.69 (5281)
>= 51	45.01 (415)	47.81 (526)	48.15 (574)	47.31 (591)	47.15 (672)	47.18 (2778)
>=75	5.53 (51)	5.36 (59)	5.12 (61)	5.36 (67)	5.33 (76)	5.33 (314)
Total Number of Observations	922	1100	1192	1249	1425	5888

Note: Figures in brackets represent the number of observations

Table 6.1: Benchmark Model of Discretionary Accruals

Variable	abs_DA_ Francis	res_Francis_ TA	pos_DA_ Francis	neg_DA_Francis
Intercept	0.0561***	0.0042	0.0317***	-0.0394***
post_cg	490.0065	0.0012	0.006	0.0015
fii_shr	0.0236	-0.0048	-0.0103	0.0482
banks_fi_shr	-0.0694***	0.0463***	-0.0171	0.072***
leverage	0.0384***	-0.0318***	0.0028	-0.057***
pb	-0.0002	0.0001	0	0.0003
Nrsize	-0.0037	0.0024	0.0005	0.0094
beta	0.0018	-0.0023*	0	0.0011
cfo_ta	-0.0107	-0.7358***	-0.5330***	-0.5434***
l1caccr_ta	-0.0166	0.0227**	-0.0083	0.0192
roa	-0.2466***	0.8019***	0.5423***	0.5921***
proa	-0.1312***	0.0800***	-0.0719**	0.1352***
vareps	0.0005	-0.0005	-0.001	0.0002
abs_roa	0.3924***	-0.3622***	-0.1317*	-0.3906***
abs_roa_gr	0.0052**	0.0021	0.0063	-0.0012
Obs	5465	5465	2949	2516
Adj R-Sq	0.1276	0.5416	0.4458	0.4734
Pr > F	0	0	0	0

Note: ***, **, and * denote levels of statistical significance at the 1%, 5% and 10% level, respectively.

Table 6.2: Effect of Promoter Control on Discretionary Accruals

Variable	abs_DA_Francis	res_Francis_TA	pos_DA_Francis	neg_DA_Francis
intercept	0.0820***	0.0029	0.0434***	-0.0517***
prom_control	-0.1303***	-0.0054	-0.0546	0.0497
prom_control_sq	0.1308***	0.0137	0.0519	-0.042
fii_shr	0.0189	0.0008	-0.0158	0.0541
banks_fi_shr	-0.0665***	0.0518***	-0.0187	0.0761***
leverage	0.0388***	-0.0319***	0.0029	-0.0581***
pb	-0.0002	0.0001	0	0.0003
nsize	-0.0047	0.0015	0.0006	0.0091
beta	0.0018	-0.0022	-0.0001	0.0012
cfo_ta	-0.0101	-0.7364***	-0.5315***	-0.5433 ***
l1caccr_ta	-0.0173	0.0231**	-0.0089	0.0196
roa	-0.2396***	0.8021***	0.5429***	0.5885***
Proa	-0.1325***	0.0779**	-0.0722**	0.1329***
Vareps	0.0005	-0.0005	-0.001	0.0002
abs_roa	0.3888***	-0.3627***	-0.1313*	-0.3896***
abs_roa_gr	0.0051**	0.0021	0.0062	-0.0011
post_cg49	0.0068	0.0013	0.0061	0.0014
Industries dummies	included	included	included	included
Time dummies	included	included	included	included
Obs	5465	5465	2949	2516
Adj R-Sq	0.1304	0.5416	0.4461	0.4734
Pr > F	0	0	0	0

Note: ***, **, and * denote levels of statistical significance at the 1%, 5% and 10% level, respectively.

Table 6.3: Effect of Group Affiliation on Discretionary Accruals

Variable	abs_DA_ Francis	res_Francis_ TA	pos_DA_ Francis	neg_DA_Francis
intercept	0.0531***	0.004	0.0307**	-0.0383***
group	0.0077**	0.0006	0.0026	-0.0028
fii_shr	0.0198	-0.0051	-0.0116	0.0496
banks_fi_shr	-0.0763***	0.0458***	-0.0196	0.0750 ***
leverage	0.0372***	-0.0319***	0.0025	-0.0569***
pb	-0.0002	0.0001	0	0.0003
nsize	-0.0102	0.0019	-0.0016	0.0119
beta	0.0017	-0.0023*	-0.0001	0.0011
cfo_ta	-0.0116	-0.7359***	-0.5330***	-0.5427***
l1caccr_ta	-0.016	0.0227**	-0.0083	0.0188
roa	-0.2451***	0.8020***	0.5423***	0.5911***
proa	-0.1272***	0.0804***	-0.0703*	0.1342 ***
vareps	0.0003	-0.0005	-0.001	0.0003
abs_roa	0.3949***	-0.3620***	-0.1309*	-0.3917***
abs_roa_gr	0.0052**	0.0021	0.0063	-0.0012
post_cg49	0.0051	0.0011	0.0056	0.0019
Industry dummies	included	included	included	included
Time dummies	included	included	included	included
Obs	5465	5465	2949	2516
Adj R-Sq	0.1284	0.5416	0.4457	0.4733
Pr > F	0	0	0	0

Note: ***, **, and * denote levels of statistical significance at the 1%, 5% and 10% level, respectively.

**Table 7.1: Effect of Promoter Control on Discretionary Accruals
- Group Affiliated Firms**

Variable	abs_DA_ Francis	res_Francis_ TA	pos_DA_ Francis	neg_DA_Francis
intercept	0.1034***	-0.0041	0.0456***	-0.0444**
prom_control	-0.2033**	-0.0271	-0.1089	0.0258
prom_control_sq	0.1909***	0.0175	0.0746	-0.0231
fii_shr	0.0212	-0.0306	-0.0222	-0.0014
banks_fi_shr	-0.0761***	0.0486**	-0.0253	0.0627**
leverage	0.0493**	-0.0273*	0.0184	-0.0582***
pb	-0.0001	0	0.0001	0.0001
nsize	-0.0042	-0.0077	-0.0102	-0.004
beta	0.0023	0.003	0.0021	-0.0013
cfo_ta	0.0346	-0.8312***	-0.5588***	-0.6899***
l1caccr_ta	-0.0132	0.0138	-0.0171	0.0047
roa	-0.2832**	1.0042***	0.7607***	0.7887***
proa	-0.0960**	0.1088***	0.0071	0.1365***
vareps	0.0004	-0.0015*	-0.0014	-0.0011
abs_roa	0.4211***	-0.4316***	-0.2800*	-0.4383***
abs_roa_gr	0.0069	0.0089*	0.0135	0.0023
post_cg49	0.0125*	0.0105	0.0142**	0.0053
Industries dummies	included	included	included	included
Time dummies	included	included	included	included
Obs	2766	2766	1452	1314
Adj R-Sq	0.1496	0.6175	0.453	0.61
Pr > F	0	0	0	0

Note: ***, **, and * denote levels of statistical significance at the 1%, 5% and 10% level, respectively.

Table 7.2: Effect of Promoter Control on Discretionary Accruals
- Standalone Firms

Variable	abs_DA_ Francis	res_Francis_ TA	pos_DA_ Francis	neg_DA_Francis
intercept	0.0690***	0.0055	0.0395***	-0.0572***
prom_control	-0.0772*	0.0129	-0.0009	0.0674
prom_control_sq	0.0764*	0.0088	0.0144	-0.0469
fii_shr	-0.0184	0.0595	-0.0076	0.2104***
banks_fi_shr	-0.0968***	0.0466	-0.0354	0.1488***
leverage	0.0221*	-0.0334***	-0.0131	-0.0517***
pb	-0.0010**	0.0004	0.0002	0.0010***
nsize	-0.0134	0.0099	0.0102	0.0251*
beta	0.0008	-0.0025	-0.0006	0.0073
cfo_ta	-0.0492	-0.6635***	-0.5189***	-0.4501***
l1caccr_ta	-0.02	0.0243	-0.0064	0.0252
roa	-0.2129***	0.6930***	0.4394***	0.4864***
proa	-0.1476**	0.053	-0.1243 **	0.118**
vareps	0.0012	0.0016	-0.0017	0.0021
abs_roa	0.3796***	-0.3438***	-0.0659	-0.3865***
abs_roa_gr	0.0036**	-0.0018	0.0014	-0.002
post_cg49	-0.0055	-0.0022	-0.0039	0.006
Industries dummies	included	included	included	included
Time dummies	included	included	included	included
Obs	2699	2699	1497	1202
Adj R-Sq	0.144	0.4876	0.5019	0.4056
Pr > F	0	0	0	0

Note: ***, **, and * denote levels of statistical significance at the 1%, 5% and 10% level, respectively.

Table 8: Group Effect by Promoter Control Thresholds

	Promoter control thresholds					
	Low (<20%)	High (≥20%)	Low (<26%)	High (≥26%)	Low (<51%)	High (≥51%)
Abs_DA_Francis	0.0064	0.0059**	0.0103	0.0058*	0.0083*	0.0109***
Res_Francis_TA	-0.018	0.0001	- 0.0009	0.0009	0.0048	-0.0032
Pos_DA_Francis	- 0.0119	0.0016	0.0046	0.0017	0.0075	-0.0004
Neg_DA_Francis	- 0.0022	-0.0028	0.0015	-0.0001	0.0001	-0.0072

Note: ***, **, and * denote levels of statistical significance at the 1%, 5% and 10% level, respectively.

Table 9.1: Summary Statistics: Opacity Analysis

Variable	All Firms		By Ownership group		
	N	Mean	Standalones	Group Affiliated	t-diff
abs_DA_Francis	2743	0.8111	0.0808	0.0815	-0.19
prom_control	2909	0.4998	0.4922	0.5079	-2.39
fii_shr	2909	0.0151	0.0056	0.0252	-11.8
banks_fi_shr	2909	0.0358	0.0167	0.0562	-17.69
opaque_1	2909	0.2585	0.2975	0.2172	7.39
opaque_2	2909	0.1860	0.1968	0.1748	3.00
nrsize	2909	0.4644	-0.0827	0.1828	-31.07
leverage	2909	0.3238	0.2864	0.3634	-8.66
vareps	2909	0.4116	0.2005	0.6346	-8.70
pb	2909	0.8006	0.6969	0.9101	-0.57
beta	2909	1.0359	1.0069	1.0665	-4.50
cfo_ta	2821	0.0583	0.0514	0.0653	-2.86
l1caccr_ta	2870	-0.0030	0.0048	-0.0112	3.67
roa	2909	0.1258	0.1196	0.1324	-2.85
proa	2909	0.0227	0.0225	0.0231	-0.20
abs_roa	2909	0.1403	0.1390	0.1418	-0.73
abs_roa_gr	2909	0.6819	0.7567	0.6031	3.74
post_cg49	2909	0.8731	0.7992	0.9512	-12.84

Table 9.2: Effect of Ownership Opacity on Absolute Discretionary Accruals
- All Firms

Variable	abs_DA_Francis			
	Column 1	Column 2	Column 3	Column 4
intercept	0.0574***	0.0750***	0.0905***	0.0786***
opaque_1	0.0094		0.0057	
opaque_2		0.0074		0.0066
prom_control			-0.1714***	-0.0804*
prom_control ²			0.1561**	0.0716
fii_shr	0.0135	0.0677*	0.0218	0.0678*
banks_fi_shr	-0.0357	-0.0241	-0.0568*	-0.0449
leverage	0.0400**	0.0146	0.0292**	0.0042
pb	-0.0003	-0.0001	-0.0002	-0.0001
nsize	-0.002	-0.0102	-0.0057	-0.0150*
beta	-0.004	0.0003	-0.0045	0.0009
cfo_ta	-0.0689	-0.0557	-0.0772***	-0.0650***
l1caccr_ta	-0.0403**	-0.0474**	-0.0339**	-0.0413**
roa	-0.163**	-0.1808***	-0.1496***	-0.1780***
proa	-0.0747	-0.1206**	-0.0647**	-0.1029**
vareps	-0.001	-0.0007	-0.0004	-0.0003
abs_roa	0.3355***	0.3050***	0.3203***	0.3020***
abs_roa_gr	0.0079	0.0028	0.0084***	0.0028*
Post_cg49	0.0042	0.0008	0.0036	0.0043
Industry dummies	included	included	included	included
Time dummies	included	included	included	included
Obs	2736	2386	2736	2386
Adj R-Sq	0.1323	0.1409	0.0951	0.0754
Pr > F	0	0	0	0

Note: ***, **, and * denote levels of statistical significance at the 1%, 5% and 10% level, respectively.

**Table 9.3: Effect of Ownership Opacity on Absolute Discretionary Accruals:
Group Affiliated and Standalone Firms**

Variable	abs_DA_Francis		abs_DA_Francis	
	Group affiliated	Standalones	Group affiliated	Standalones
Intercept	0.1059	0.0783	0.1196	0.0551
opaque_1	0.0164*	-0.0011		
opaque_2			0.0111	0.0034
Prom_control	-0.2286**	-0.0988	-0.1903**	-0.0179
Prom_control ²	0.1961**	0.0965	0.1634**	0.0211
fii_shr	-0.0002	0.0431	0.0516	0.0631
banks_fi_shr	-0.0663*	-0.0769	-0.0528	-0.0620
leverage	0.0278**	0.0243	-0.0076	0.0148
pb	-0.0002	-0.0014	0.0004	-0.0014**
nrsiz	-0.0155	-0.0058	-0.0182	-0.0195
beta	-0.0057	-0.0048	0.0019	0.0005
cfo_ta	-0.1166***	-0.0548	-0.0535	-0.0742**
l1cacr_ta	-0.0752**	0.0049	-0.0697	-0.0126
roa	0.0061	-0.2360	-0.1386	-0.2113***
proa	-0.0557	-0.0599	-0.0793	-0.1179**
vareps	-0.0004	-0.0019	0.0001	-0.0027
abs_roa	0.2121***	0.3786	0.2183	0.3737***
abs_roa_gr	0.0143***	0.0040	0.0023	0.0036
Post_cg49	0.0037	0.0027	0.0027	0.0032
Industry dummies	included	included	included	included
Time dummies	included	included	included	included
Obs	1376	1360	1176	1210
Adj R-Sq	0.1096	0.0971	0.0529	0.0923
Pr > F	0	0	0	0

Note: ***, **, and * denote levels of statistical significance at the 1%, 5% and 10% level, respectively.

Table 10.1: Effect of Non-Disclosure (OPAQUE_1) on Absolute Discretionary Accruals - Group Affiliated Firms

	Promoter Control Thresholds					
	Low (<20%)	High (≥20%)	Low (<26%)	High (≥26%)	Low (<51%)	High (≥51%)
intercept	0.1985	0.0804***	-0.1125	0.0797***	0.2972	0.0771***
opaque_1	-0.1467	0.0123	0.0270	0.0110	0.0363**	-0.0035
fii_shr	0.3279	0.0408	-0.2098	0.0481	0.0011	0.0035
banks_fi_shr	-0.6938	-0.0601**	-0.2864*	-0.0443	-0.0531	0.0537*
Leverage	0.3058	-0.0007	0.1525**	-0.0015	0.0571**	-0.1078
pb	-0.3031***	-0.0001	0.0181	-0.0001	-0.0001	-0.0189**
nsize	0.2695	-0.0148	-0.0180	-0.0119	-0.0109	-0.0006**
beta	-0.1311	-0.0050	-0.0335	-0.0027	-0.0099	-0.0349
cfo_ta	0.1467	-0.0615**	-0.1289	0.0697***	-0.1713***	0.0028
l1caccr_ta	-0.1493	0.0788***	0.0769	0.0765***	-0.0639***	-0.0256***
roa	0.5675	-0.1149**	0.4539**	0.1246***	0.0819	-0.0819**
proa	0.7294	-0.0802**	-0.3420	-0.0567	-0.0725	-0.1832
vareps	-0.1568	0.0005	0.0114	0.0003	-0.0011	-0.0463
abs_roa	0.7466	0.1874	0.6073**	0.1925***	0.3056***	0.0017
abs_roa_gr	0.0633***	0.0029***	0.0638***	0.0032	0.0812***	0.2165**
post_cg49	-0.0199	-0.0013	0.0434	-0.0032	-0.0112	0.0046
Obs	36	1340	101	1275	691	685
Adj R-Sq	0.7813	0.0524	0.5960	0.0480	0.1779	0.0376
Pr > F	0	0	0	0	0	0

Note: ***, **, and * denote levels of statistical significance at the 1%, 5% and 10% level, respectively.

Table 10.2: Effect of Fragmentation (OPAQUE_2) on Absolute Discretionary Accruals - Group Affiliated Firms

	Promoter Control Thresholds				
	High# (≥20%)	Low (<26%)	High (≥26%)	Low (<51%)	High (≥51%)
intercept	0.0681	0.1139	0.0659***	0.0868**	0.0524**
opaque_2	0.0099	0.0304	0.0109	0.0003	0.0490**
fii_shr	0.0707	-0.0977	0.0744*	0.0931*	0.0274
banks_fi_shr	-0.0372	-0.2393	-0.0258	-0.0564	-0.0614
leverage	-0.0021	-0.0147	-0.0049	0.0138	-0.0249
pb	0.0005	0.0223	0.0004	0.0001	0.0015
nsize	-0.0157	0.0256	-0.0147	-0.0104	-0.0231
beta	0.0033	-0.1018	0.0063	-0.0094	0.0110
cfo_ta	-0.0160	-0.4209**	-0.0177	-0.0721**	-0.0429
l1caccr_ta	-0.0876	0.2436	-0.0830***	-0.0383	-0.0949**
roa	-0.1471	-0.0822	-0.1606***	-0.1461**	-0.1663*
proa	-0.0463	-0.6833**	-0.0123	-0.1698***	0.0343
vareps	0.0002	0.0137	0.0001	0.0001	0.0008
abs_roa	0.1885	0.5813**	0.1974**	0.2275**	0.2055**
abs_roa_gr	0.0029	-0.0004	0.0033	-0.0016	0.0093**
post_cg49	-0.0009	0.0644	-0.0021	-0.0040	0.0050
Obs	1160	65	1111	600	576
Adj R-Sq	0.0386	0.3288	0.0323	0.0554	0.0546
Pr > F	0	0	0	0	0

Note: # There were too few observations to estimate the “LOW” regression for the 20 % threshold.

***, **, and * denote levels of statistical significance at the 1%, 5% and 10% level, respectively.